Suggested problems: 1-51 odd

Problem 1: Compute the following iterated integrals.

(a) $\int_0^3 \int_0^4 (4x + 3y)\ dx\ dy$

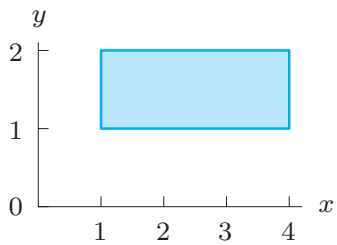
(b) $\int_0^2 \int_0^3 (x^2 + y^2)\ dy\ dx$

(c) $\int_0^1 \int_0^1 ye^{xy}\ dx\ dy$

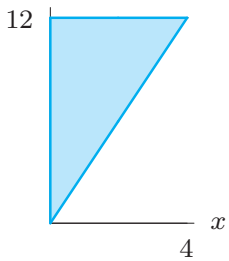
Problem 2: A building is 8 meters wide and 16 meters long. It has a flat roof that is 12 meters high at one corner, and 10 meters high at each of the adjacent corners. What is the volume of the building?

Problem 3: For each part below, write $\int_R f\ dA$ as an iterated integral of the shaded region R.

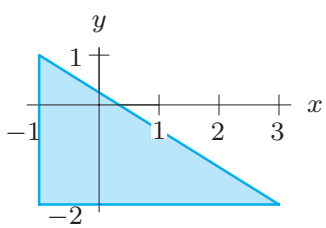
(a)



(b)



(c)



Problem 4: Compute the following iterated integrals. Sketch the region R of integration.

(a) $\displaystyle\int_0^2 \int_0^x e^{x^2}\ dy\ dx$

(b) $\displaystyle\int_1^4 \int_{\sqrt{y}}^y x^2 y^3\ dx\ dy$

(c) $\int_1^5 \int_x^{2x} \sin(x)\ dy\ dx$

Problem 5: Compute the following iterated integrals by reversing the order of integration.

(a) $\int_0^2 \int_y^2 e^{x^2}\ dx\ dy$

(b) $\int_0^1 \int_{x^2}^1 \sqrt{y} \sin(y)\ dydx$